

Rapid TCA

Spencer Butler
May 2025

Overview

Rapid TCA is a Jupyter Notebook designed to generate on-demand trade cost analysis (TCA) for US equity orders when unavailable through the OMS or a third-party. It reads fill-level execution data, enriches it with market data, calculates and displays trading performance metrics.

The purpose is to provide traders and investment managers with immediate, high-level feedback on execution quality without replacing formal post-trade TCA reporting.

Background

Rapid TCA focuses on the post-trade reporting subset of TCA. Post-trade reporting evaluates how well a trade was executed after the fact. It compares execution price and timing against benchmarks such as arrival price and volume-weighted average price (VWAP).

The goal of post-trade reporting is to evaluate trading costs, identify areas for improvement, and ensure best execution. Reports are typically produced through dedicated software and analyzed by trading managers, compliance, and the client. Investment managers use post-trade reporting to guide broker selection.

Features

- Completion Rate
- Percentage of Volume
- Implementation Shortfall
- Interval VWAP
- Venue Analysis

Tools

- Python (Pandas, Matplotlib)
- Bloomberg API (via xbbg)
- Jupyter Notebook

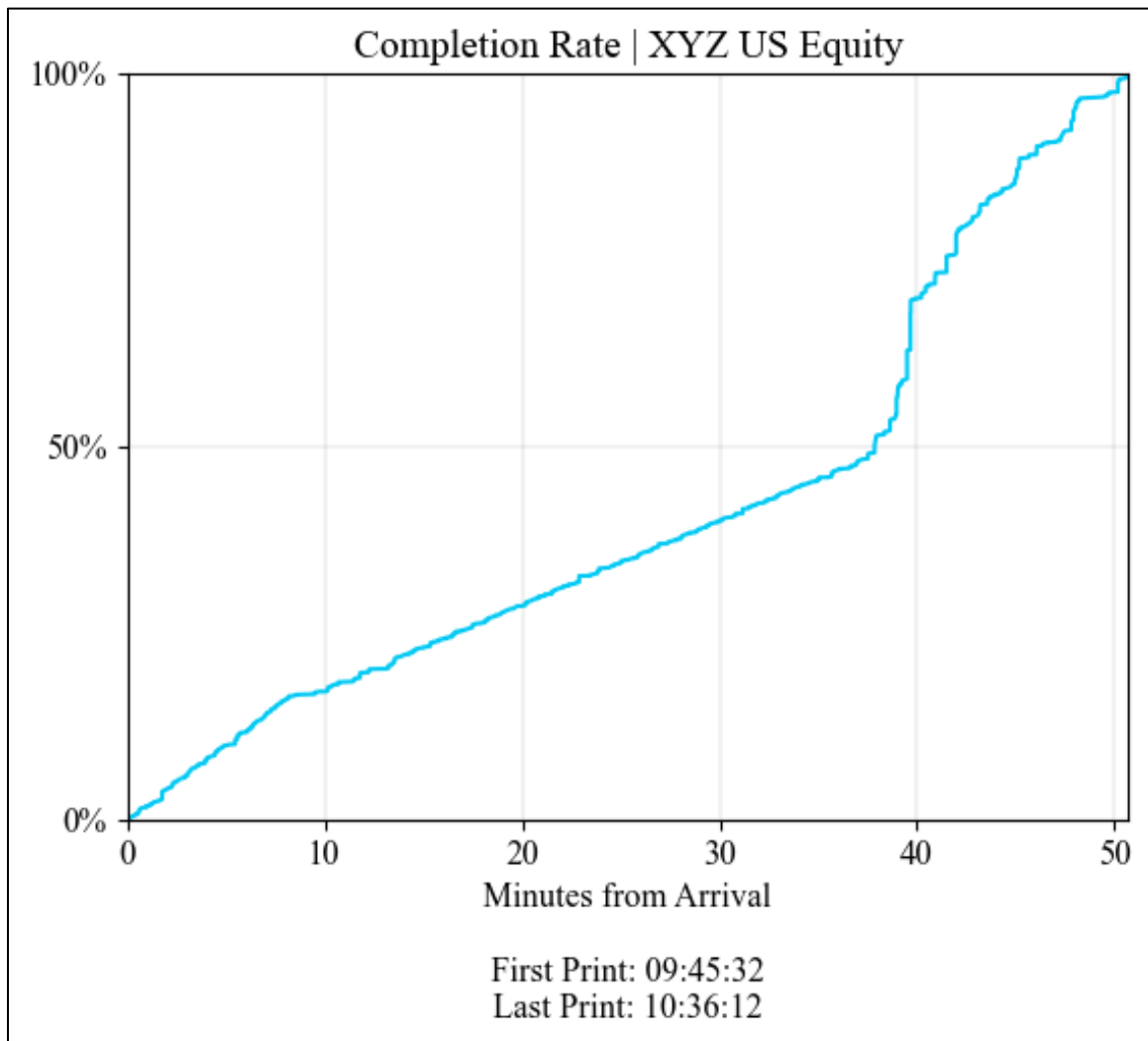
Disclaimers

Example data is random and does not represent any real-world trade. Full documentation is available on GitHub.¹ This does not constitute investment advice.

¹ <https://github.com/spencerbutler888/rapid-tca>

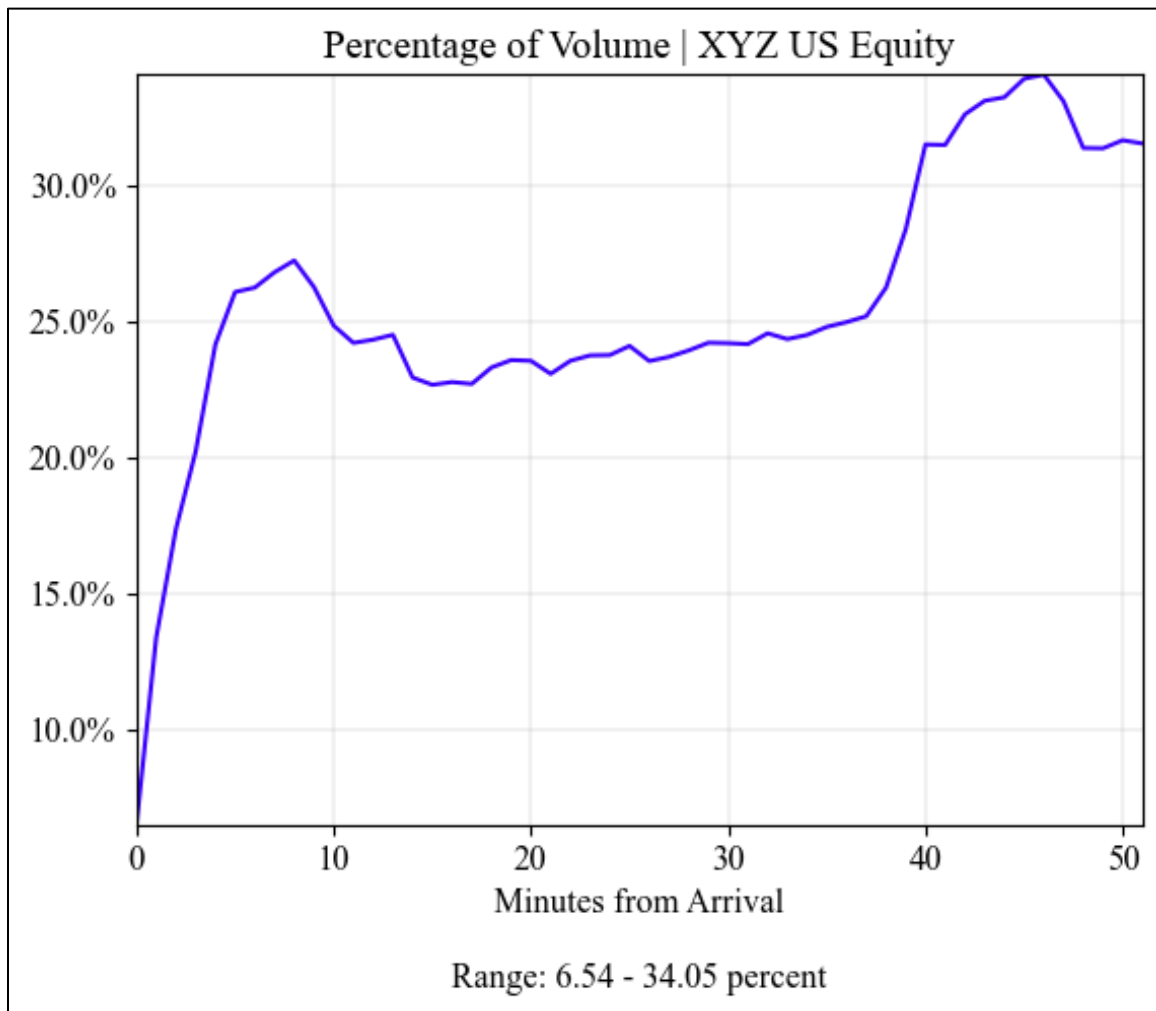
Completion Rate

Percentage of the total order quantity filled, measured in 1-minute buckets.



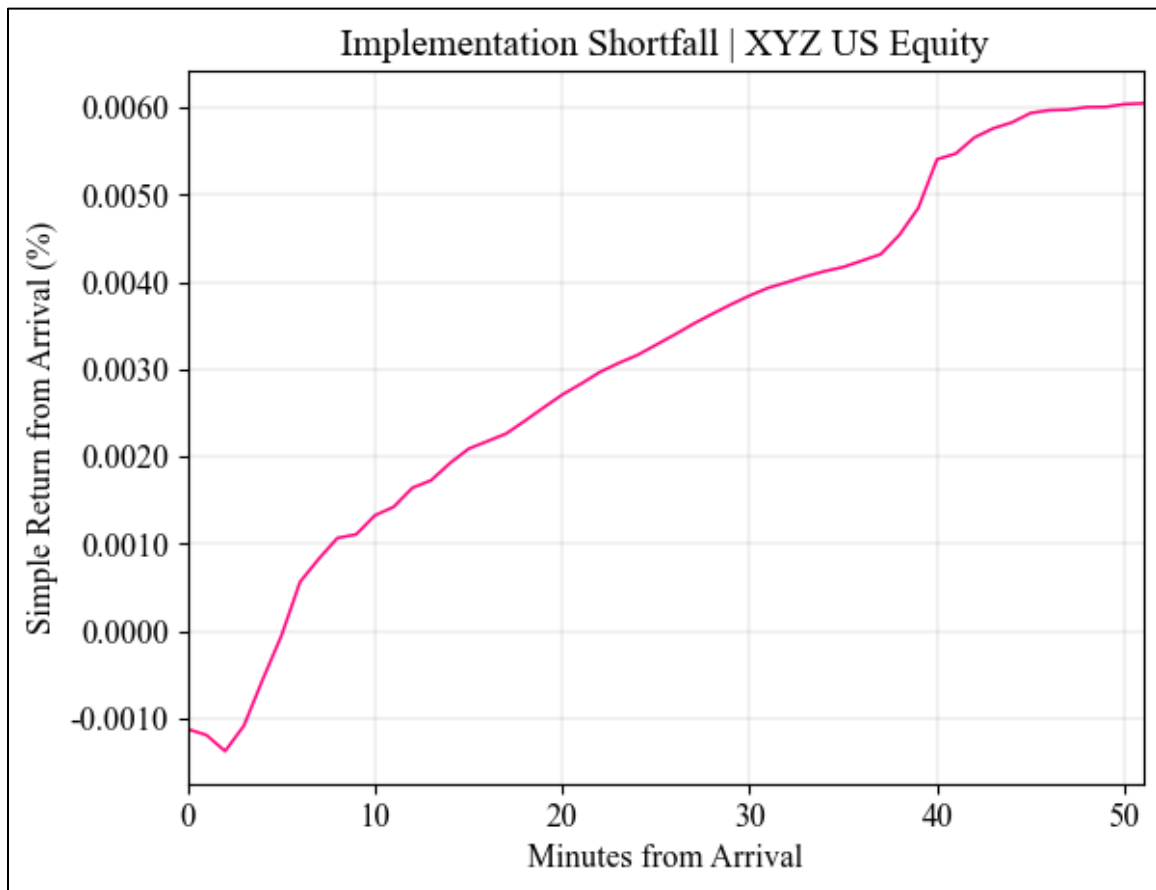
Percentage of Volume

Client shares executed as a percentage of total market volume, measured in 1-minute buckets.



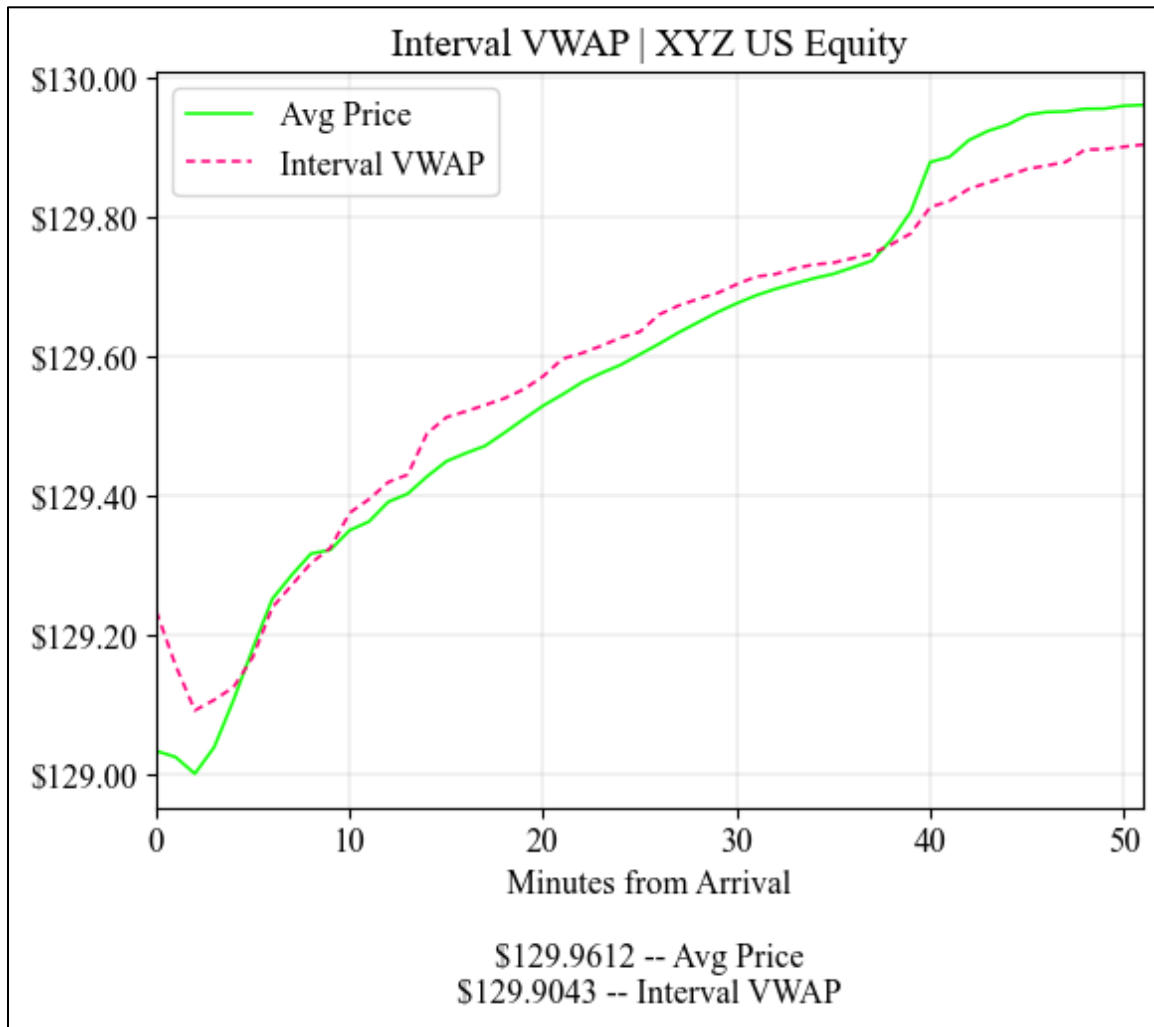
Implementation Shortfall

Simple return of the rolling, volume-weighted average execution price relative to the arrival price.



Interval VWAP

Rolling volume-weighted average execution price compared to market IVWAP, measured in 1-minute intervals.



Venue Analysis

Execution quantity broken down by venue and venue type.

